FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

FIITJEE **JEE (Main), 2016**

ALL INDIA INTEGRATED TEST SERIES

HALF COURSE TEST – V

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 02 and 09 to 30)** contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (03 to 08) contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 marks** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

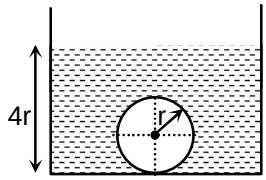
Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. When the temperature of a gas, filled in a closed vessel, is increased by 1^0C its pressure increases by 0.4 %. The initial temperature of the gas is,
(A) 250 K (B) 2500 K
(C) 250^0C (D) 25^0C

2. A string of linear mass density 0.8 kg/m is stretched to a tension 500 N. The mean power required to maintain a travelling wave of amplitude of 10 mm and wavelength 0.5 m is
(A) 70 W (B) 85.3 W
(C) 98.7W (D) 110 W

3. A spherical metal ball of radius 'r' is lying at the bottom of a stationary container containing liquid of density ρ as shown in the figure. Find the force exerted on the upper hemispherical portion of the sphere due to gauge pressure (P_0 = atmospheric pressure).

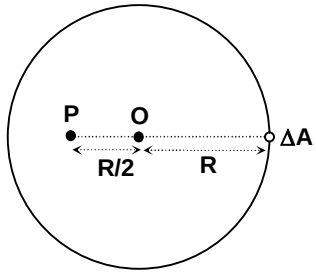


(A) $\frac{\pi r^2}{3}\left[3P_0 + 7r\rho g\right]$ (B) $\frac{\pi r^2}{2}\left[3P_0 + 7r\rho g\right]$
(C) $\pi r^2\left[3P_0 + 7r\rho g\right]$ (D) $2\pi r^2\left[3P_0 + 7r\rho g\right]$

4. A spring mass system executes damped harmonic oscillations given by the equation
$y = Ae^{-\frac{bt}{2m}}\sin(\omega' t + \phi)$ where $\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$,
where the symbols have their usual meanings.
If a 2 kg mass (m) is attached to a spring of force constant(K) 1250 N/m, the period of the oscillation is (π/12)s. The damping constant 'b' has the value.
(A) 9.8 kg/s (B) 2.8 kg/s
(C) 98 kg/s (D) 28 kg/s

Space for rough work

5. A small section of area 'ΔA' is removed from a uniform spherical shell with surface mass density σ and radius 'R' as shown in the figure. Find the magnitude of gravitational field intensity at point P due to the remaining mass.



(A) $\frac{4\sigma\Delta AG}{9R^2}$ (B) $\frac{4\sigma\Delta AG}{R^2}$

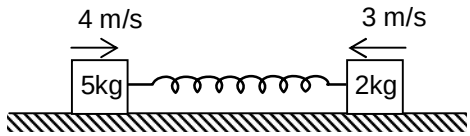
(C) $\frac{\sigma\Delta AG}{R^2}$ (D) zero

6. A horizontal spring mass system is executing SHM with time period of 4 sec. At time t = 0, it is at mean position. Find the minimum time after which its potential energy becomes three times of kinetic energy.

(A) 1 sec (B) 1/2 sec

(C) 1/3 sec (D) 2/3 sec

7. Two blocks connected with the spring of force constant 100 N/m are given velocities $4\hat{i}$ m/s and $-3\hat{i}$ m/s when the spring is in natural length as shown in figure



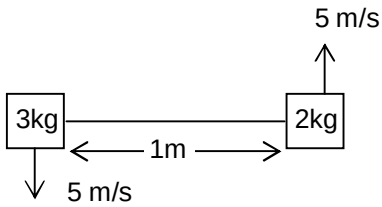
(A) The velocity of 2kg block is maximum when 5 kg block is at instantaneous rest.

(B) The maximum and minimum velocities of 2kg block is $7\hat{i}$ m/s and $-3\hat{i}$ m/s respectively.

(C) The maximum and minimum velocities of 5kg block is 4 m/s and zero respectively.

(D) All of the above

8. Two blocks of masses 2kg and 3kg are connected by string of length 1m. At any instant the velocities of block of mass 2kg and 3kg is 5 m/s in opposite direction and perpendicular to the length of string and is also parallel to horizontal table.



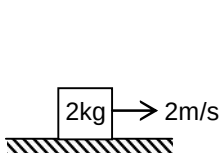
(A) Tension in the string is 110 N

(B) Tension in the string is 120 N

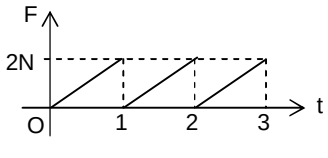
(C) Velocity of centre of mass is 2 m/s

(D) Velocity of centre of mass is 5 m/s

9. A block of mass 2kg starts moving at t = 0 with speed 2m/s on a smooth horizontal surface. A horizontal force F is applied in the direction of velocity which varies with time shown in figure (b) the speed of particle at t = 3 seconds (g = 10 m/s^2)



(a)



(b)

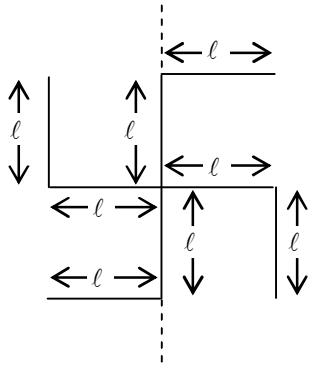
(A) 2.5 m/s (B) 3.5 m/s

(C) 4.5 m/s (D) none of these

Space for rough work

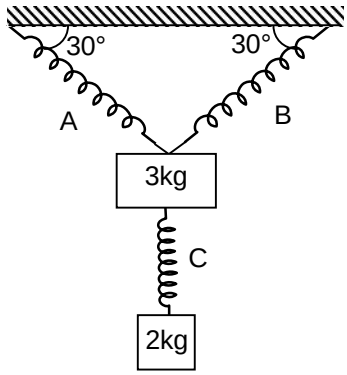
10. Two objects of mass 3kg and 2kg move along x and y axis with speed 4 m/s and 3 m/s respectively on the horizontal smooth table. After collision the bodies stick together. Then
(A) Heat generated in the process is 15 J
(B) Heat generated in the process is 18 J
(C) Direction of motion with x-axis after collision is 60°.
(D) Direction of motion with x-axis after collision is $\tan^{-1}\left(\frac{1}{3}\right)$.

11. In the shown figure the linear mass density of the rod is λ and dimension are shown in figure. The moment of inertia about the dashed axis is



(A) $\frac{4}{3}\lambda\ell^3$
(B) $\frac{8}{3}\lambda\ell^3$
(C) $\frac{10}{3}\lambda\ell^3$
(D) $4\lambda\ell^3$

12. A block of mass 3kg is connected with two identical massless springs A and B and a block of mass 2kg is hanging with a third identical spring C as shown in figure then the magnitude of acceleration of 3kg and 2kg block just after cutting the spring A



(A) $\frac{25}{3}$ m/s^2 and zero
(B) $\frac{50}{3}$ m/s^{2}and zero
(C) $\frac{50}{3}$ m/s^2, 10 m/s^2
(D) 20 m/s^2, zero m/s^2

13. Two particles of masses 1kg and 2kg are moving with constant velocities 2m/s$(\hat{i})$ and 5m/s$(\hat{i})$ respectively and crosses the y-axis simultaneously at t = 0 sec and are moving on a smooth horizontal xy-plane. The separation between the two particles is 10 meter at t =0. The angular momentum of 2kg particle with respect to 1kg particle at t = 5 sec is
(A) 20 N-m-sec (B) 40 N-m-sec
(C) 60 N-m-sec (D) 80 N-m-sec

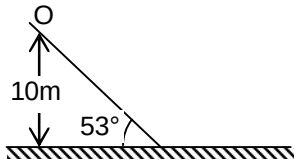
Space for rough work

14. A particle moves along an arbitrary path. If $\vec{v}$ and $\vec{a}$ are the instantaneous velocity and acceleration vectors of the particle, then the ratio of magnitude of tangential and centripetal acceleration is

(A) $\frac{|\vec{a}\cdot\vec{v}|}{|\vec{a}\times\vec{v}|}$ (B) $\frac{|\vec{a}\times\vec{v}|}{|\vec{a}\cdot\vec{v}|}$

(C) $\frac{d\vec{v}/dt}{d|\vec{v}|/dt}$ (D) None of these

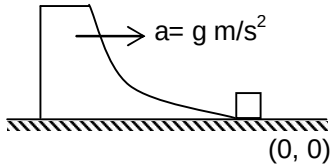
15. A ball is released from rest at a height of 10m on an inclined plane of inclination 53°. If collision is in elastic then find the horizontal velocity of the ball



(A) $10\sqrt{2}$ (B) $6\sqrt{2}$

(C) $12\sqrt{2}$ (D) $14\sqrt{2}$

16. A wedge whose curved surface is parabolic in shape and has equation $x^2 = 4y$ starts accelerating with acceleration g m/s^2 when a block is at the bottom of wedge and is located at (0, 0). The maximum height attained by the block (assume the curved surface is sufficiently long)



(A) 1m (B) 2m

(C) 4m (D) 8m

17. The ratio of magnitude of work done in moving slowly a block of mass m on a rough inclined surface of inclination $\theta = 45°(\tan\theta > \mu)$ from bottom to top and from top to bottom is 5. The coefficient of friction between the block and surface is

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$

(C) $\frac{2}{5}$ (D) $\frac{4}{5}$

18. Four identical particles of mass m are projected along the plane towards the centre of a square from the corners of the square simultaneously with constant speed V_0. Then

(A) If the two spheres comes to rest after collision then the other two must also come to rest after collision.

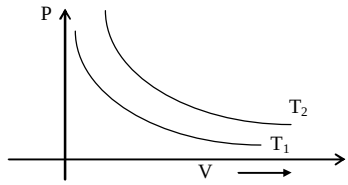
(B) If the three spheres comes to rest after collision then the fourth one must also stop after collision.

(C) If all comes to rest after collision then loss of energy is $4m V_0^2$.

(D) If all comes to rest after collision then it will violate the law of conservation of momentum.

Space for rough work

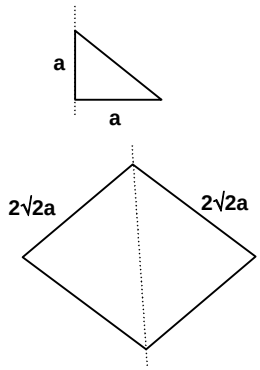
19. Figure shows the pressure P versus volume V graphs for a certain mass of an ideal gas at two constant temperatures T_1 and T_2.



Which of the following option is correct?
(A) $T_1 = T_2$
(B) $T_1 > T_2$
(C) $T_1 < T_2$
(D) no inference can be drawn due to insufficient information.

20. Two identical vessels A and B contain mass m and 2m of same gas respectively. The gases in the vessels are heated keeping their volumes constant and equal. The Temperature-Pressure curve for mass 2m makes angle α with T-axis and that for mass m makes an angle β with T-axis then
(A) $\tan\alpha = \tan\beta$ (B) $\tan\alpha = 2\tan\beta$
(C) $\tan\beta = 2\tan\alpha$ (D) None of these

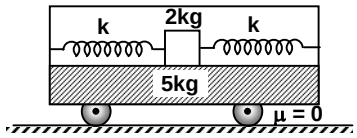
21. If the moment of inertia of a isosceles right angle plate is I about an axis shown in figure. Then moment of inertia of a square plate of same material and thickness shown in figure about the given axis is
(A) 30I (B) 32I
(C) 64I (D) 90I



22. Two simple pendulum of length ℓ and 16 ℓ are released from the same phase together. They will be at the same phase after a minimum time.
(A) $\frac{8\pi}{3}\sqrt{\frac{\ell}{g}}$ (B) $\frac{\pi}{3}\sqrt{\frac{\ell}{g}}$
(C) 2s (D) none of these

Space for rough work

23. Two springs, each of spring constant k = 100 N/m are attached to a block of mass 2 kg as shown in figure. The block can slide smoothly along horizontal platform clamped to the opposite walls of trolley of mass 5 kg. The block is slightly displaced and then released. The period of oscillation is (in seconds). (all surfaces are smooth).



(A) $T = 2\pi\sqrt{\frac{7}{1000}}$ (B) $T = 2\pi\sqrt{\frac{1}{140}}$

(C) $T = 2\pi\sqrt{\frac{1}{20}}$ (D) $T = 2\pi\sqrt{\frac{49}{100}}$

24. Two particles are executing SHM of equal amplitude along parallel lines with same time period. Their velocity vectors are always oppositely directed to each other. The minimum phase difference between the two particles is

(A) 0° (B) 90°

(C) 180° (D) 270°

25. Which of the following statements is **incorrect?**

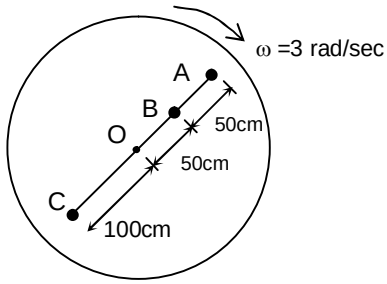
(A) Work done is zero if the point of application of force does not move.

(B) Total work done by static friction is zero in rolling without slipping for the complete system.

(C) Work done by friction must be zero in pure rolling for the rolling body.

(D) For an isolated system momentum of the system is conserved for any type of collision.

26. Three small spheres A, B and C of masses 1kg, 2kg and 3kg are rotating with a circular disc with angular velocity 3 radian/sec and are connected with strings to O as shown in figure. Coefficient of friction between spheres A, B, C and disc is 0.4, 0.3 and 0.2 respectively.



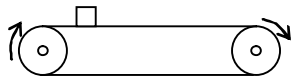
(A) Friction force acting on sphere B is 6 N.

(B) Tension in the string connecting sphere A and B is 9 N

(C) Tension in the string connecting sphere C and O is 8 N

(D) Data insufficient

27. A block of mass 2kg is kept gently on a moving tape of a machine in which the rotating cylinders have angular speed of 20 rad/sec and there radius is 20cm. If coefficient of friction between the tape and block is 0.5 and there is no slipping between tape and cylinder. Then (take g = 10 m/s^2)



(A) The distance travelled by the block before relative motion with tape ceases is 1.6 m

(B) Work done by friction on the block is 8 N

(C) Work done by friction on the block is – 16 N

(D) Linear velocity of tape is 40 m/s

Space for rough work

28. A rod of mass m kg and length ℓ meter is hinged about its end A and is vertical initially. Now the end A is accelerated horizontally with acceleration $a_0 = g$ m/s^2. The hinge reaction when the rod becomes horizontal is



(A) $R_x = \frac{4mg}{3}N, R_y = \frac{mg}{3}N$ (B) $R_x = \frac{mg}{3}N, R_y = \frac{mg}{4}N$

(C) $R_x = \frac{mg}{4}N, R_y = \frac{mg}{3}N$ (D) $R_x = 4mgN, R_y = \frac{mg}{4}N$

29. The temperature at which the speed of sound wave in helium gas is same as that in hydrogen gas at 27°C, is

(A) 504°C (B) 45°C

(C) 327°C (D) 231°C

30. A triangular wave pulse on a taut string travels in positive x-direction with speed v. The tension in the string is F, and linear mass density of string is μ. At t = 0, the shape of pulse is given by

$$y(x,0) = \begin{cases} 0 & \text{if} \quad x < -L \\ \frac{h(L+x)}{L} & \text{for} \quad -L < x < 0 \\ \frac{h(L-x)}{L} & \text{for} \quad 0 < x < L \\ 0 & \text{for} \quad x > L \end{cases}$$

choose the correct statement.

(A) Magnitude of instantaneous power is zero for $-L < (x - vt) < 0$

(B) Magnitude of instantaneous power is $Fv\left(\frac{h}{L}\right)^2$ for $(x - vt) < -L$

(C) Magnitude of instantaneous power is $Fv\left(\frac{h}{L}\right)^2$ for $(x - vt) > L$

(D) Magnitude of instantaneous power is $Fv\left(\frac{h}{L}\right)^2$ for $0 < (x - vt) < L$

Space for rough work

Chemistry

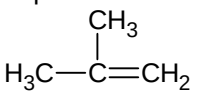
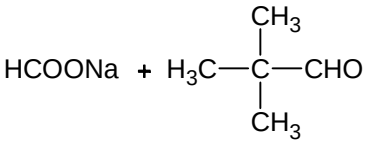
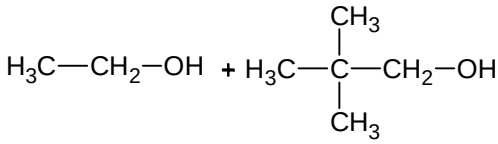
PART - 1

SECTION – A

Straight Objective Type

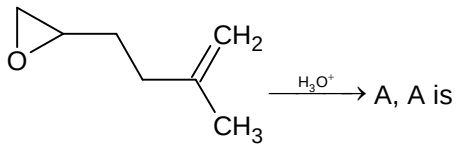
This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which only one is correct.

1. The velocity of electron in the second orbit of He^+ will be
   (A) 2.18×10^6 m / s (B) 1.09×10^6 m / s
   (C) 3.8×10^6 m / s (D) 6.8×10^{-6} m / s

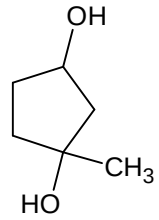
2. A mixture of $(CH_3)_3C - CHO$ and HCHO on heating with aqueous KOH solution will give
   (A) HCOONa + $(CH_3)_3C - CH_2 - OH$
   (B) 
   (C) 
   (D) 

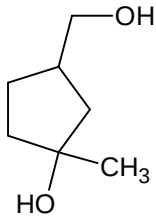
3. For the reaction:
   $R - X + OH^- \longrightarrow ROH + X^-$
   The rate of reaction is given as follows.
   Rate = 4.74×10^{-5} [R – X] [OH^-] + 0.24×10^{-5} [RX].
   What percentage of R – X will react by SN_2 path if [OH^-] = 10^{-3} molar
   (A) 1.93 (B) 4.74
   (C) 2.37 (D) 4.9

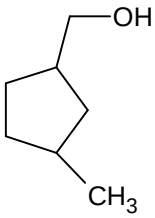
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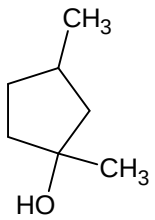
4. 

A, A is

(A) 

(B) 

(C) 

(D) 

5. If the radius of Bohr's is x, then de Broglie wavelength of electron in 3rd orbit is nearly
(A) $4\pi x$ (B) $6\pi x$
(C) $8\pi x$ (D) $\pi\frac{x}{3}$

6. Which of the following reagent solution can be used to distinguish between methanoic acid & ethanoic acid?
(A) Tollen's reagent (B) $FeCl_3$ solution
(C) $FeSO_4/H_2O_2$ (D) Na_2CO_3 solution

7. Si-F bond is stronger than C-F bond due to
(A) Larger size of silicon atom
(B) Larger electronegativity difference between Si & F
(C) The presence of $p\pi$ - $d\pi$ bonding
(D) Overlapping of P – P atomic orbital

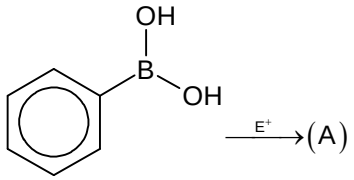
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8. Which of the following species have the bond order same an N_2?
(A) CN^- (B) OH^-
(C) NO^- (D) CO^+

9. In FCC lattice of NaCl structure, if the diameter of Na^+ is X & radius of Cl^- is Y, then bond length of NaCl in the crystal is
(A) 2x + 2y (B) x + y
(C) $\frac{x}{2}+\frac{y}{2}$ (D) $\frac{x}{3}+\frac{x}{2}$

10. CsCl has bcc structure with Cs^+ at the centre and Cl^- ion at each corner. If rCs^+ = 1.69 $\overset{0}{A}$ & rCs^- = 1.81 $\overset{0}{A}$. What is the edge lenght 'a' of each cube?
(A) $\frac{5}{\sqrt{2}}$ (B) $\frac{6}{\sqrt{3}}$
(C) $\frac{7}{\sqrt{3}}$ (D) $\frac{4}{\sqrt{3}}$

11. In the metallurgy of iron, when lime stone is added to the blast furnace, the calcium ion ends up in
(A) slag (B) gangue
(C) metallic calcium (D) calcium carbonate

12. 

In the product (A), the electrophile will be attached to the which position w.r.t $-B(OH)_2$ group
(A) ortho (B) para
(C) meta (D) ipso

13. Find total number of product as a result of given reaction:
$CH_3-CH=CH-CH_2-COOAg \xrightarrow{Br_2/CCl_4/\Delta}$
(A) 3 (B) 4
(C) 5 (D) 6

14. The aqueous solution of each of the following salt is coloured except
(A) $TiCl_4$ (B) $FeCl_3$
(C) $CuCl_2$ (D) $MnCl_2$

Space for rough work

15. Which of the following salt will not produce black ppt on passing $H_2S(g)$ through their aqueous salt solution in acidic medium?
(A) $CuSO_4$ (B) $PbCl_2$
(C) $CdSO_4$ (D) $NiCl_2$

16. A current of 2A passing for 1.93×10^4 sec through a molten Tin salt depositing 23.8 g Tin. The oxidation state of Tin in the salt is (given atomic number of Tin = 119)
(A) 2 (B) 4
(C) 1 (D) 3

17. What volume strength of 200 ml H_2O_2 required for complete reaction of 7.9 g of $KMnO_4$ in basic medium?
(A) 1.86 (B) 11.2
(C) 1.4 (D) 5.6

18. The molecular weight of an acid is 82. In a titration, 100 ml of solution of this acid containing 39 gm of acid per litre were completly neutralized by 95 cc of aqueous NaOH containing 40 gm NaOH per litre. What is the basicity of the acid?
(A) 1 (B) 2
(C) 3 (D) 4

19. Two sample of HI each of 5 gm were taken separately in two vessels of volume 5 & 10 litre respectively at 27°C. The extent of dissociation of HI will be
(A) more in 5 litre vessel (B) more in 10 litre vessel
(C) equal in both vessel (D) nota

20. The ratio of equivalent weight of oxidant & reductant in
$C_2H_5OH + OH^- + I_2 \longrightarrow CHI_3 + HCO_2^- + H_2O + I^-$
(A) 1:2 (B) 2:1
(C) 2:3 (D) 3:2

21. What is the pH of 10^{-3} M solution of NH_4OH ($K_b = 1.85 \times 10^{-5}$ M) at 25°C?
(A) 10.75 (B) 10.14
(C) 9.65 (D) 9.10

22. Which of the following compound would you expect to be molecular?
(A) N_2O (B) PCl_5
(C) $CaCl_2$ (D) both (A) & (B)

Space for rough work

23. Which one of the following shows maximum paramagnetic character?

(A) $\left[Cr(H_2O)_6\right]^{3+}$ (B) $\left[Fe(CN)_6\right]^{4-}$

(C) $\left[Fe(CN)_6\right]^{3-}$ (D) $\left[Cu(H_2O)_6\right]^{2+}$

24. Boron has two isotopes $^{10}_{5}B$ and $^{11}_{5}B$. If the atomic weight of boron is 10.81, the ratio of $^{10}_{5}B$ and $^{11}_{5}B$ in nature is

(A) 19/81 (B) 20/53

(C) 15/16 (D) 10/11

25. Which of the following is non stoichiometric and metal deficient?

(A) FeO (B) Fe_3O_4

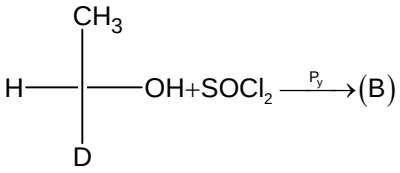
(C) Fe_2O_3 (D) all of these

26. Percentage of gold in 22 carat gold is

(A) 15 (B) 85

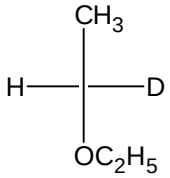
(C) 92.6 (D) 37.5

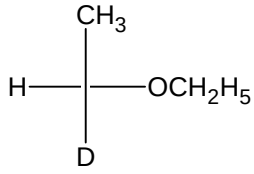
27. $CH_3-CH_2-OH+Na\longrightarrow(A)$

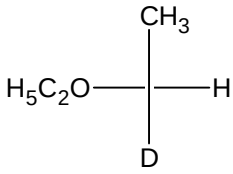


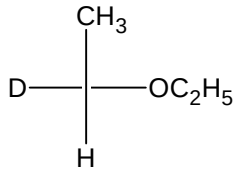
$A+B\longrightarrow(C)$

Find (C).

(A) 

(B) 

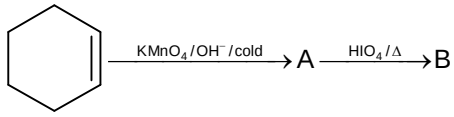
(C) 

(D) 

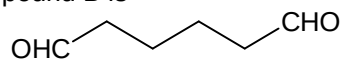
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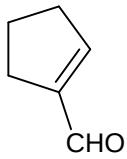
28. Potassium 40 decay to Argon – 40 with a half life of 1.27×10^9 years. What is the age of rock in which the mass ratio of ^{40}Ar to ^{40}K is 3.6?

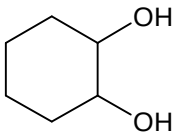
(A) 3.2×10^9 years
(B) 4.2×10^9 years
(C) 2.8×10^9 years
(D) 6.4×10^8 years

29. 

Compound B is

(A) 

(B) 

(C) 

(D) None

30. The number of moles of $Cr_2O_7^{2-}$ needed to oxidise 0.136 equivalent of $N_2H_5^+$ by the reaction:

$N_2H_5^+ + Cr_2O_7^{2-} \longrightarrow N_2 + Cr^{3+} + H_2O$

is

(A) 0.136
(B) 0.272
(C) 0.816
(D) 0.0227

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

*This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct*

1. If $N = \left(7 + 4\sqrt{3}\right)^n = P + B$ $(n \in N)$ where P is the integral part of at N and B is the positive proper fraction then the value of (1 – B)(P + B)
   (A) 1 (B) 2
   (C) 3 (D) 4.

2. If {x} denotes the fractional part of 'x' then find the value of $\left\{\frac{3^{2n}}{8}\right\}$ where $n \in N$.
   (A) $\frac{1}{2}$ (B) $\frac{1}{4}$
   (C) $\frac{1}{8}$ (D) $\frac{1}{16}$.

3. The 8^m term in the expansion of $(1+2x)^{-1/2}$
   (A) $(1)^7 \frac{(1.3.5...13)x^7}{7!}$ (B) $(-1)^7 \frac{(1.3.5...13)x^7}{7!}$
   (C) $(-1)^6 \frac{(1.3.5...13)x^6}{6!}$ (D) None of these.

4. If A_0, A_1, A_2, A_3, A_4 and A_5 be the consecutive vertices of a regular hexagon inscribed in a unit circle. Then find the product of length of A_0A_1, A_0A_2, and A_0A_4.
   (A) 1 (B) 3
   (C) 5 (D) 7

Space for rough work

5. Number of positive real value of x such that x, [x] and {x} are in H.P. (where [x] and {x} denotes the greatest integer and fraction part of x) is equal
(A) 1 (B) 2
(C) 0 (D) 3.

6. From the points on the circle $x^2 + y^2 = 4$ tangents are drawn to the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$. If the locus of middle points of the chord of contact is the curve $a(bx^2 + cy^2) = (dx^2 + ey^2)^2$, then the value of $\left[\frac{a+b+c+d+e}{5}\right]$ where ([.] $\rightarrow$ G.I.P.)
(A) 4 (B) 5
(C) 6 (D) 9.

7. Let $a_1, a_2, a_3 \ldots a_{201}$ are in G.P. with $a_{101} = 25$ and $\sum_{i=1}^{201} a_i = 625$. Then one value of $\sum_{i=1}^{201} \frac{1}{a_i}$ is equal to
(A) 5 (B) 25
(C) 125 (D) 1.

8. If $x = \sec\theta - \cos\theta$ and $y = \sec^3\theta - \cos^3\theta$ then the value of $\left(\frac{dy}{dx}\right)^2$ at x = 0 is
(A) 0 (B) 2
(C) 4 (D) 9.

9. $\frac{x}{a}+\frac{y}{b}=1$ is a variable line, where $\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{c^2}$ (c is a constant). Locus of the foot of perpendicular drawn from origin to the variable line is
(A) $x^2 - y^2 = c^2$ (B) $x^2 + y^2 = c^2$
(C) $-x^2 + y^2 = c^2$ (D) $-x^2 - y^2 = c^2$

10. If the locus of the middle point of the chords of the parabola $y^2 = 4ax$ which subtends angle α at the vertex is $(32 + y^2 - 4x)^2 \tan^2\alpha = 64(8x - y^2)$ then the value of a is
(A) 1 (B) 2
(C) 3 (D) 4.

Space for rough work

11. The sum of real values of k for which the cubic $x^3 - kx + k - 1 = 0$ has exactly two distinct real solution

(A) $\frac{13}{4}$ (B) $\frac{15}{4}$

(C) $\frac{3}{4}$ (D) $\frac{11}{4}$

12. In triangle, if $\Delta = a^2 - (b - c)^2$ then tanA is equal to

(A) $\frac{7}{15}$ (B) $\frac{8}{15}$

(C) $\frac{11}{15}$ (D) $\frac{13}{15}$.

13. The value of the expression $\sum_{n=1}^{\infty} \frac{\tan(2n-1)\pi}{\frac{4}{2^{n+1}}}$ is

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{1}{6}$ (D) $\frac{2}{3}$.

14. Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and let m(b) be one minimum value of f(x). As b varies, the range of m(b) is

(A) [0, 1] (B) $\left(0, \frac{1}{2}\right]$

(C) $\left[\frac{1}{2}, 1\right]$ (D) $(0,1]$.

15. A line y = x + 2 is drawn on the co-ordinate plane. This line is rotated by 90º clockwise about the point (0, 2). A line y = – 2x + 10 is drawn and area of this triangle is Δ, then the value $[\Delta]$ is ($[] \rightarrow$ is G.I.F.)

(A) 128 (B) 64

(C) 21 (D) 20.

Space for rough work

16. Number of integers satisfying the inequalities
$\sqrt{\log_3 x - 1} + \dfrac{\frac{1}{2}\log_3 x^3}{\log_3 \frac{1}{3}} + 2 > 0$, is
(A) 5 (B) 6
(C) 7 (D) 8.

17. If the equation $\left|\left|x+3\right|-2\right| = p$, where p is a constant integer has exactly three distinct solution then the number of integral values of p, is
(A) 0 (B) 1
(C) 2 (D) 4

18. Let $A = \{x : (x+2)(x-4) \le 0\}$, and $B = \{x : x^2 - ax + 4 \le 0\}$ if $A \cup B = A$, then the largest integral value of a is
(A) 3 (B) 4
(C) 5 (D) 6.

19. If $\cos\theta_1 = \dfrac{2\cos\theta_2 - 1}{2 - \cos\theta_2}$ where $\theta_1, \theta_2 \in (0, \pi)$, then the value of $\dfrac{\tan^2\frac{\theta_1}{2}}{\tan^2\frac{\theta_2}{2}}$ is equal to
(A) 1 (B) 2
(C) 3 (D) 4.

20. Let $P(x) = x^5 + x^2 + 1$ have roots x_1, x_2, x_3, x_4 and x_5 $g(x) = x^2 - 2$, then the value of $g(x_1) \cdot g(x_2) \cdot g(x_3) \cdot g(x_4) \cdot g(x_5) - 30g(x_1x_2x_3x_4x_5)$ is
(A) 1 (B) 2
(C) 4 (D) 7.

21. If $a + b + c = 0$ and $a^2 + b^2 + c^2 = 1$ then the value of $2(a^4 + b^4 + c^4)$ is
(A) 1 (B) 2
(C) 3 (D) 3.

Space for rough work

22. If the value of $\alpha \in [0, 2\pi]$ such that the inequality $\sin\left(\frac{\pi}{3} + x\right) + \sin(\alpha + x) \geq 0$ is true for all that number x, is equal to $\frac{p\pi}{q}$ where p and q are relatively prime positive integers, then the value of (p + q) is
(A) 3 (B) 5
(C) 7 (D) 9

23. $f(x) = x^3 + px^2 + qx + 6$, where $p, q \in R$ if $f'(x)$ is negative in largest possible interval $\left(\frac{-5}{3}, -1\right)$, then the value of (p + q) is
(A) 3 (B) 5
(C) 7 (D) 9

24. In a ΔABC, the tangent of half the difference of two angles is one third the tangent of half the the sum of same angles. The ratio of the sides opposite to the angles.
(A) 2 : 1 (B) 1 : 2
(C) 2 : 4 (D) 4 : 2.

25. If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = 3\pi$ then compute the value of
$$x^{2004} + y^{2004} + z^{2004} + \frac{6}{x^{2003} + y^{2003} + z^{2003}}$$
(A) 0 (B) 1
(C) 2 (D) none of these.

Space for rough work

26. Area of the parallelogram formed by the lines y = mx, y = mx + 1, y = nx and y = nx + 1 equals

(A) $|m + n|(m - n)^2$ (B) $\frac{2}{|m+n|}$

(C) $\frac{1}{|m-n|}$ (D) $\frac{1}{|m+n|}$.

27. Let PQ and RS be tangents at the extremities of the diameter PR of a circle of radius r, If PS and RQ intersect at a point x on the circumference of the circle, then 2r equals.

(A) $\sqrt{PQ \,.\, RS}$ (B) $\frac{(PQ+RS)}{2}$

(C) $\frac{2PQ \,.\, RS}{(PQ+RS)}$ (D) $\sqrt{\frac{(PQ^2+RS^2)}{2}}$.

28. If a > 2b > 0 then positive value of m for which $y = mx - b\sqrt{1+m^2}$ is a common tangent to $x^2 + y^2 = b^2$ and $(x - a)^2 + y^2 = b^2$ is

(A) $\frac{2b}{\sqrt{a^2-4b^2}}$ (B) $\frac{\sqrt{a^2-4b^2}}{2b}$

(C) $\frac{2b}{a-2b}$ (D) $\frac{b}{a-2}$.

29. The locus of the orthocentre of the triangle formed by the lines
(1 + p)x – py + p(1 + p) = 0
(1 + q)x – qy + q(1 + q) = 0 and y = 0, where $p \neq q$ is
(A) a hyperbola (B) a parabola
(C) an ellipse (D) a straight line.

30. Let $a_1\ a_2 \ldots a_{10}$ be in A.P. $h_1\ h_2 \ldots h_{10}$ be in H.P. if $a_1 = h_1 = 2$ and $a_{10} = h_{10} = 3$, then $a_4\ h_7$is
(A) 2 (B) 3
(C) 5 (D) 6

Space for rough work